Picomenu 2.4.0 User Guide

A big thanks to Brian Apps for creating the Picoface proof of concept, without this Picomenu would not have been possible.

Note on release 2.4.0

Release 2.4.0 adds no additional features to the standard pico. This release adds additional features to the newly supported Pico W - these features are specific to wifi and bluetooth.

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Setting up Picomenu

You need the following to to setup and use Picomenu

- A Picoface (with a pico or pico w)
- And SD card (formatted to exFAT or FAT32)
- The latest release in .uf2 format.
- 48.rom and 128.rom (or 128-0.rom and 128-1.rom). Don't ask me for these - I cannot provide them.

You need to flash the Raspberry Pi Pico with the .uf2 (there are lots of guides on the internet on how to do this) and insert the SD card into the Piceface SD card slot.

Plug it into your Spectrum expansion port and switch on the spectrum.

The NMI Button

The NMI button (top right of the Picoface) has four functions:

1) A short press takes you into the NMI menu.

2) Holding it for three seconds resets the Spectrum.

3.) Holding it down for one second takes you to the Save snapshot screen

3) A double press takes a quick screen grab and takes you to the screen grabs category screen.

Some of the above settings can be changed in the options menu.

Get Started

Before doing anything you can ask Picomenu to set up your SD card.

Choose T-Tools, and then D-Set up the SD card. It will list the things it will create on your SD card. Press D and it will confirm it has written those files. If you ever delete a file or folder and need it re-created, the Set up SD card can do this. It will only write what it cannot find,

so it will not overwrite any of your existing files or folders that you previously used SD card setup to create.

The Main Menu

If you are on the latest release when you press the NMI button you will see this menu.



Here is a summary of what each item does, and I'll expand on some of these later in the guide.

B- Browse games	Takes you to the game and file browser
S - save snapshot	Takes to the save snapshot menu
R - Reload:	A quick reload of the last game played
E - Recently played	A rolling list of the last ten games played
C- Change Rom	Takes you to the ROM change menu
P - Pokes	Takes you to the Pokes menu
4 - Reset to 48k	Resets to 48k
1 - Reset to 128k	Resets to 128k

H - Help	Help text, along with a diagnostic page.
I - Game Instructions	Takes you to the V-Screen (see further information below)
V - Virtual Tape Player	Takes you to the virtual tape player screen
T - Tools	A selection of helpful tools and options
A - Autosave	Take you to the autosave menu
X - Exit	Takes you back to the Spectrum

B-Browse Games

This is the main game/file browser and the functions are all listed at the bottom of the screen.

Here you can navigate folders, copy, move, delete files and folders and load games.

The file formats that can loaded are

.tap
.tzx
.szx
.z80
.sna

These files are all displayed by default.



The following functions can also be performed.

L-All	This will hide or unhide hidden files and folders. You can set the default setting in the option menu.
S-Card and I-Internal Flash	The Disk section will highlight in blue what is currently selected - press S-SD Card or I-Internal Flash to select which you want to use (note the internal flash is only 2mb, so functions are limited on the internal flash).
O-Info	Will show additional file information - Type, size and mode. You can set the default setting in the option menu.
T-Tape	Insert a .tap or .tzx into the Virtual Tape Player
V-View	View Game information (see below)
A-Fav	This adds a game to the Favourities folder
F-Find	Search for a file or folder. Press F again to clear the search.

The V-View Screen

If you press V and then choose a game you will be taken to the View Screen



From here you can do the following

Enter	Load the game
F-Full	Show the screen grab in full screen (you can press Enter on the full screen to load the game)
D-Del	Delete any screen grabs associated with this game (that have the same file name)
M-Edit	Here you add game details, like Software House, Year, etc. (You can also manually edit the instructions.txt file from a PC). You can also add your own notes.
X-Bk	Takes you back to the Browse games screen.
N or P	Cycles through screens

If you are viewing a .tap or .tzx file then on the second page of the V-Screen you will be given an option to press C to convert the tape to a .z80.

If you press C you will be given three options:

A-Auto (will try a 128k conversion first and fall back to 48k if that fails
4-48k (a 48k .z80)

1-128k (a 128k .z80)

After you choose an option the conversion will start. As part of the conversion it will take a screen grab of the loading screen (see G-Screen Grab). Once complete it will place the .z80 in the same file as the .tap or .tzx it was created from.

S-Save Snapshot

If you are running a game and you enter the S-Save Snapshot menu you will see this screen.

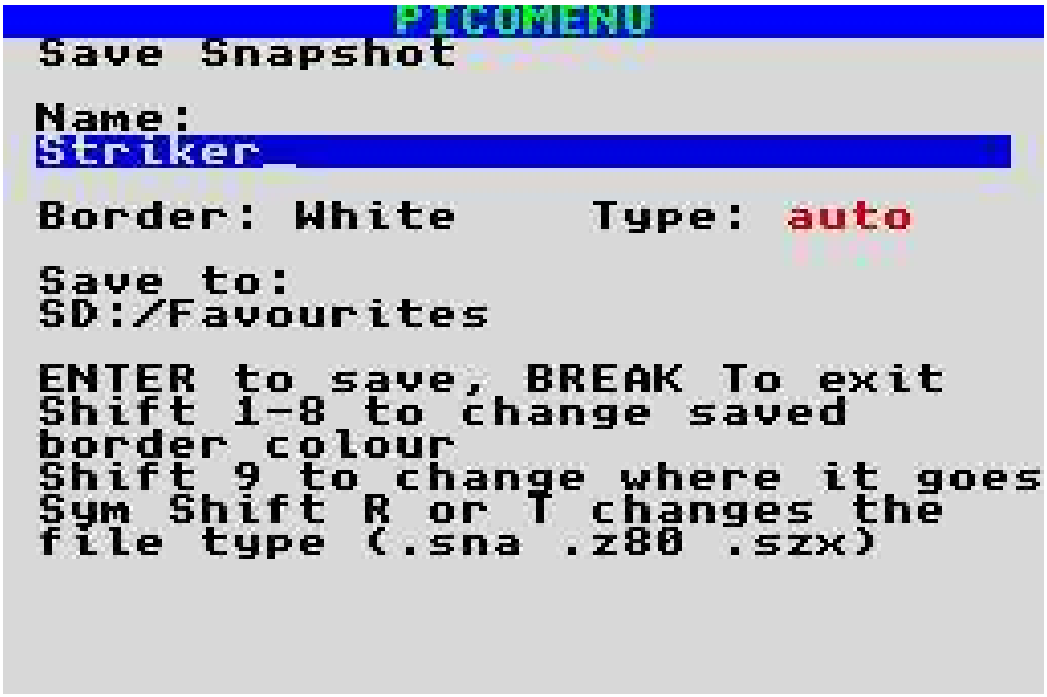
Here you can create a snapshot to be used as a save game, ten for each game, these are stored in /Saved Games/<Game name>.

You can use L-Label to give your save game a label, for example you can label the slot as second screen, or level one, etc and V-View will view the screen you saved on..



If you load a game and one or more of the save slots is populated then you see this screen and will be asked if you want to carry on from a slot or start the game. You can also view each screen from the snap shot and it will say last to the last file that was saved.

If you press S-Save snapshot (own name). You will be greeted with this screen where you can take a manual snap shot.



R-Reload:

This is a quick reload of the last game you played

E-Recently played

A rolling list of the last ten games you have played.

C-Change ROM

If you place .rom files in the following locations then you can browse them from here.

- Root of the SD Card
- /ROMS on the SD Card
- Root of the internal flash

1 - Internal ROM is the machine's own internal ROM. You may need to switch back this after you have run a test rom.

W-Writable on this screen makes the ROM area above address 63 writable, for programs that patch it.

P-Poke

You can enter pokes manually from this screen. Alternatively, if you place a ,pok file in the /Poke List directory Picomenu can then list the pokes using the L-Poke list screen.

For example, if you have Commando.tap as the game you load then Commando.pok needs to be in the /Poke list folder for it to recognise it.

4-Reset to 48k

Resets the machine to 48k basic screen

1-Reset to 128k

Resets the machine to 128k menu screen

O-Options

The options screen allows you to customise several features of Picomenu.If you wish you can also edit the picomenu.ini file found in the root of the SD card

Theme	Choose between light or dark
Show_all	This relates to the default choice for the Browse Games screen. On shows all files/folders including hidden ones, off only shows viewable ones. Last uses the setting you last used.
Details	This relates to the additional file details O-Info. It uses the same setting as Show_all.
Sort	How you want to sort files.
Folders	How you want to sort folders.
Start_drive	Choose whether to start with the SD or Flash as the viewable drive.
Start Folder	Choose which folder to start in when you power on. / is root. If you browse to a folder in the Browse game screen and then press x to exit, you can then choose this folder from here as the default one to use.
Save_type	Choose the default save type, or leave on auto
Quick_save	When you use quick save what do you want picomenu to do if a file already exists - ask you, overwrite or create a new file.
Save_screen	Choose what happens when you press S-Save Snapshot. Does it go to the save slots screen or the screen where you can take a manual snap shot.
Slots_on_load	If you load a game and a save is in a save slot, do you want it to ask you about loading it or just enter the game

Grab-format	Screen grabs can be taken in .scr or .png or both. If you use both the V-View screen will only read one of each type to stop it repeating screens.
Autosave	Set autosave to Off, 20, 40 or 60 seconds. See A-Autosave section.
Long-press	Outside of the NMI menu you can press the NMI button for 1 second and the spectrum will reset. Use this setting to duration of the press or switch it off.
Double-press	You can double press the NMI button during a game to take a quick screen grab. You can turn this off here.
Loading-grab	The C-Conver tool (see V-View Screen) will take a loading screenshot grab when converting a .tap or .tzx to a .z80. You can switch this off here.
Waste-bin	Any file or folder you delete goes to a waste bin. If you want to permanently delete the files without using the wastebin you can switch it off there.
Level_loaders	Choose patch or off. See section on Multiload games
Lock_48k	This setting relates to multiload games, the settings are tapes, always and off.

N-Order (Screen Grab Display Order)

The next screen is the N-Grab order and relates to the screen grabs in the V-View screen. Here you can choose the order the screen grabs are displayed and turn off some categories. “From file” are screen grabs that may already be contained in a .z80, .sna or .szx. The rest of the categories are explained in G-Screen Grab.

Wifi and Bluetooth (Pico W only)

Here is where you input your network name and password and bluetooth settings.The operation of Wifi file transfer and bluetooth tape playing is explained in the T-Tools section.

Diagnostic screen

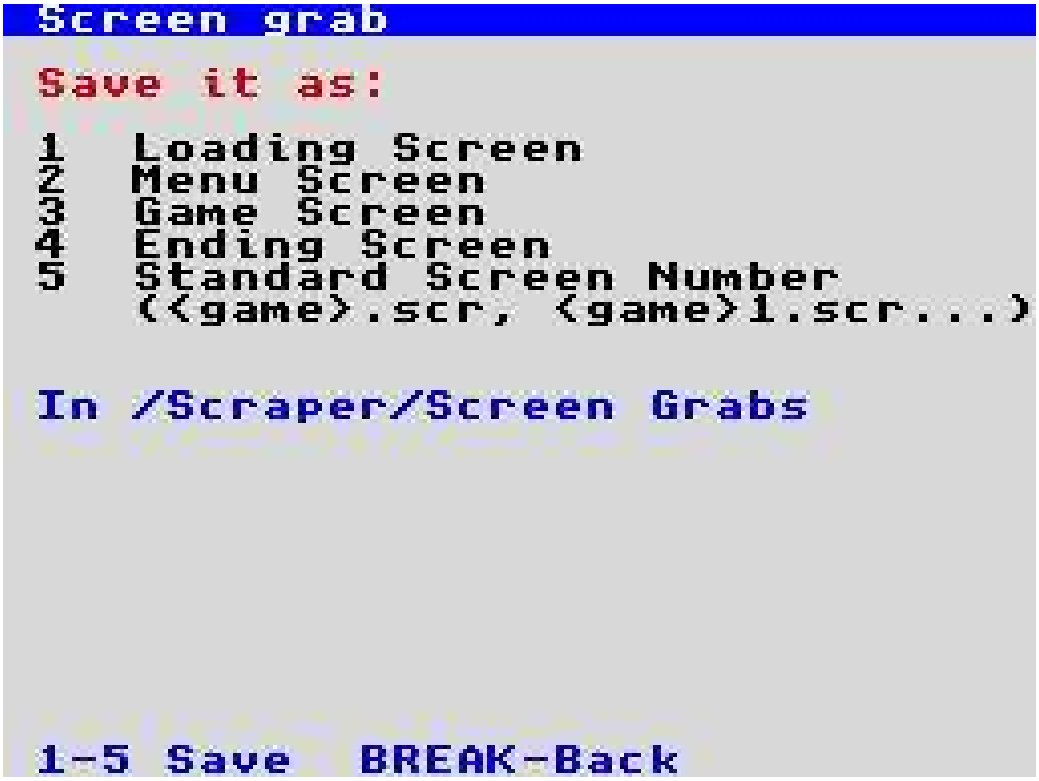
There is also a diagnostic screen (press D) this shows some basic information about the firmware and storage space.

I-Game Instructions

The same screen as the V-View screen in the Browse Games menu, but here it displays information for the game currently running.

Screen Grabs

This will take a screen grab of the current game screen. To take screen grab double press the NMI button and you will see this screen.



When you press this button you will see the screen above. You can then choose a category for your screen. It will then save a screen grab to /Scraper/Screen grabs using your games file name and the category you choose. These screens are then displayed in the V-View screen in the order you choose.

You can create multiple screens in the same category, as Picomenu will simply use a numbering system to avoid overwriting files.

V-Virtual Tape Player

If you insert a .tap or .txz in using T-Tape in the Browse Games screen then you can view the blocks of that tape here. You can skip backwards and forwards to different parts of the tape.

This is helpful for some multiload games.

When you insert a .tap or .txz in using T-Tape the picomenu will reset back the basic or 128k menu screen. However, the tape remains inserted and you can load from here in the traditional way by typing load"" on a 48k Spectrum or starting the tape loader on a 128k model.

You can add a label to any of the blocks by pressing L-Label. Here you can type a short description. For example "start of multiload levels". If you delete the label the original tape block information will be displayed.

You can also record to a tape, so you can save any programmes you have written on your spectrum.

T-Tools

There are a few tools on the Picomenu.

T-Test ROM	If you place the files below in the location indicated you can use them to test your spectrum. Testrom.rom and/or DiagROM.rom must be in one of the following locations: - - Root of the flash - Root of the SD card - /ROMS
D-Set up SD card	This is covered at the start of the user guide. This tool sets up the SD card with the correct folders and files.
U-Update	Use this to update the Picomenu firmware. Place a valid .uf2 file on the roof of the SD card and use this tool to update. You are advised to delete the old picomenu.ini file and use D-Set up SD card to create a new one where a release includes new features.
W-Waste Bin	This is where you can retrieve any accidentally deleted files from. You can also empty the waste bin from here.
C-Batch Convert	If you have a lot of .tap or .tzx files you want to convert you can put them all in a folder and ask the batch converter to convert them all automatically. It will place the created .z80 in a folder inside the folder you pointed it at called "converted" It will also provide a summary of what it has converted. You can also convert your .z80 files back to .taps from here.
F - File transfer (wifi)	You first need to add your wifi network name and password (see options). Once Picomenu has connected you can use your PC or phone browser to transfer files, or rename and erase them. There are three addresses you can use which are listed on the File Transfer screen.

B- Bluetooth tape	If you have a bluetooth mod on your spectrum then you can pair Picomenu with the bluetooth mod and load .tap and .tzx files in realtime. The first time you use this feature you will need to pair a Bluetooth device. You can use T-Tape on the Browse game screen to insert a tape. Use T-Tape, select a .tap or .tzx and then press B-Bluetooth. You will then exit back to the either the 48k basic screen or the 128k menu. From here press the NMI button choose T-Tools, then B-Bluetooth tape. Press P to play. You will drop back into the either 48k basic or the 128k menu. Either type LOAD"" in 48k basic or choose the Tape Loader on 128k machines and the .tap or .tzx will load in realtime. This is an experimental feature at this stage.

A-Autosave

You can turn on or set the Autosave to off, or 20, 40 or 60 seconds in the options screen. Autosave will take a save game automatically using the duration set.

When the save is taken there will be a short pause during the game.

When you start a game you will also be asked if you want to play with autosaves on or off.

You can access the Autosave using the A-Autosaves screen. Here you can do the following:-

1-5	Select autosave to load
V	View an autosaves save screen
M	Move an autosave to a save game slot
C	Clear the autosaves
O	Turn off autosaves

If you start a game with autosave turned off, then you can go to the A-Autosaves screen and use O to turn them on.

Multiload Games

Multiload is experimental at this stage

Picomenu has the ability to hand each part over from the tape file at once, as the game asks for it, so you never wait for a cassette.

48K multi-load games

Loading the tape directly. In B - Browse Games, choose the .tap or .tzx. The Spectrum resets into 48 BASIC and types LOAD "" automatically. From then on, each time the game asks for a level, Picomenu finds it on the tape (by name or in order) and hands it over at once.

- At the end of the tape it starts again from the beginning, as if rewound, so a game that searches for a level by name always finds it.
- This works for games whose levels load with the ROM's tape loader (Gauntlet, Gauntlet II, etc), and for games with their own copy of it (see below).

Converting first. If the main game uses a turbo or protected loader, convert it using B - Browse Games.

- The converter notes where the .tap or .tzx had got to when the game stopped loading.
- You can then load the .z80 and Picomenu remembers the tape position in the .tap or .tzx and can then load the next multiload level.
- Make sure you use 4:48k not A (Auto). On a 128K Spectrum Auto makes a 128K .z80, which plays the 128K version and so will not work.

Save Snapshot. A save snapshot will also remember the tape position from the .tap or .tzx

This means you can load the save later and it picks up the same point in the .tap or .tzx at the same tape block. This note on tape positions follows the save, and the .tap or .tzx so the position is still remembered even if you rename or move either.

On a 128K Spectrum. A 48K multi-load game runs with the 128 locked in 48K mode (as 48 BASIC does), so it behaves exactly as on a 48K machine. While locked, a long press of the NMI button resets into 48 BASIC; use the Spectrum's own reset button or a power off to get back to the 128 menu.

128K multi-load games

128K games load through the Spectrum's own ROMs, so the tape goes in without a reset.

1. Load the game: by placing a .tap or .tzx in the Virtual Tape Player and using the 128's Tape Loader. Alternatively use the converter (convert with C and **1 - 128K**, or A on a 128K) to create a .z80.
2. Insert the tape with the levels on using the T-Tape command.

3. If that tape has 48K levels first, wind it to the 128K ones using the Virtual Tape Player,
4. Carry on playing. Each level loads at once when the game asks for it.

Save Snapshot. A 128K .z80 made by the converter does not remember the tape, so after loading a .z80 you need to insert a .tap or .tzx into the virtual tape player. To skip that, save with **Q** (or S) once the tape is in and wound: the save remembers the tape and block, so when you load the .z80 in the future it it will automatically insert the .tap or .tzx

Games with their own level loader

Some games load their levels with their own copy of the ROM's tape loader, kept in memory (Out Run's Speedlock releases do). The Picoface only sees the ROM, so these used to wait at PRESS PLAY. Picomenu now finds that copy and makes it load from the tape file.

- It looks when you put in a .tap or .tzx in with T-Tape, when a snapshot indicates there is more tape data than just the main game and in each block that the .tap or .tzx loads.
- Nothing changes for games that don't have such a loader.
- If you eject the tape, the game's loader uses the Spectrum's own ROM loader, so a real cassette would load at the normal speed rather than the game's faster one.
- Loaders that work differently (for example ones that decrypt as they load) are not recognised; convert those games instead.

Multiload Settings

Found in **O - Options** on the NMI menu.

Level_loaders	Finds a game's own copy of the tape loader and makes it load from the .tap or .tzx file. Off leaves every game as is.
Lock_48k	On a 128 Spectrum, this locks a program in 48k mode. The tapes setting locks it in 48k mode when either a .tap or .tzx is loaded via the virtual tape player. Always locks in 48k mode for all snapshots. Off never locks in 48k mode.

Appendix A - Worked example of multiload: Out Run

Out Run loads a new stage at each fork in the road. Different releases need slightly different steps.

48K, Speedlock 3 tape (on a 48K or a 128K Spectrum)

1. Press B - Browse Games, and then V-View on the .tzx for Outrun, go to the tape page, press C, then 4 - 48K to convert it into a 48k .z80.
2. When it says Saved, load the new .z80.
3. Press 1 to start. Stage 1 loads from the tape at once, and later stages follow as you drive.

128K, Speedlock 4 two-tape release (on a 128K Spectrum)

1. Convert Tape 1 Side A in a 128K .z80, then load the .z80.
2. Press B - Browse Games, The T-Tape and choose, Tape 2 Side B (the stages). Every stage on it is 128K, so there is nothing to wind.
3. Press 1. Stage 1 loads and the race starts.
4. Optional: press Q before step 3 make a quick save and load that save next time; the stages tape goes back in by itself.

128K, the full Speedlock 3 tape (48K and 128K stages on one tape)

1. Convert the .tzx to a 128k .z80, then load the .z80.
2. Press B - Browse Games, then T-tape, and then choose the same .tzx.
3. Go into the Virtual Tape Player, go to page 2 and select the first 11kb block (this should be 5). Press X twice to exit.
4. Press 1. Stage 1 loads and the race starts.

Without step 3 the game takes the 48K stage 1 on the .tzx, which comes first on the tape and has the same stage number, and causes a crash.

Multiload Troubleshooting

Problem	Fix
The game waits at PRESS PLAY	Put the tape in with T (128K), or load the tape with its number (48K). Check level_loaders = patch.
A 48K game plays as the 128K version on a 128	Convert with 4 - 48K, not A, and keep lock_48k = tapes.

The game crashes as a level loads	The tape is at the wrong set of levels: wind it in V to the levels for your version (48K or 128K).
Loading the save doesn't put the tape back	The tape was moved outside PicoMenu (on a PC) or erased. Diagnostics (O - Options, D) says "Tape not found". Put it back, or put the tape in again with T and save once more.
The tape keeps going round	The game is searching for a level that isn't on the tape. Check you have the right tape (some releases put the levels on a second tape).
A long press goes to 48 BASIC, not the 128 menu	The 128 is locked in 48K mode for the game. Use the Spectrum's own reset button.